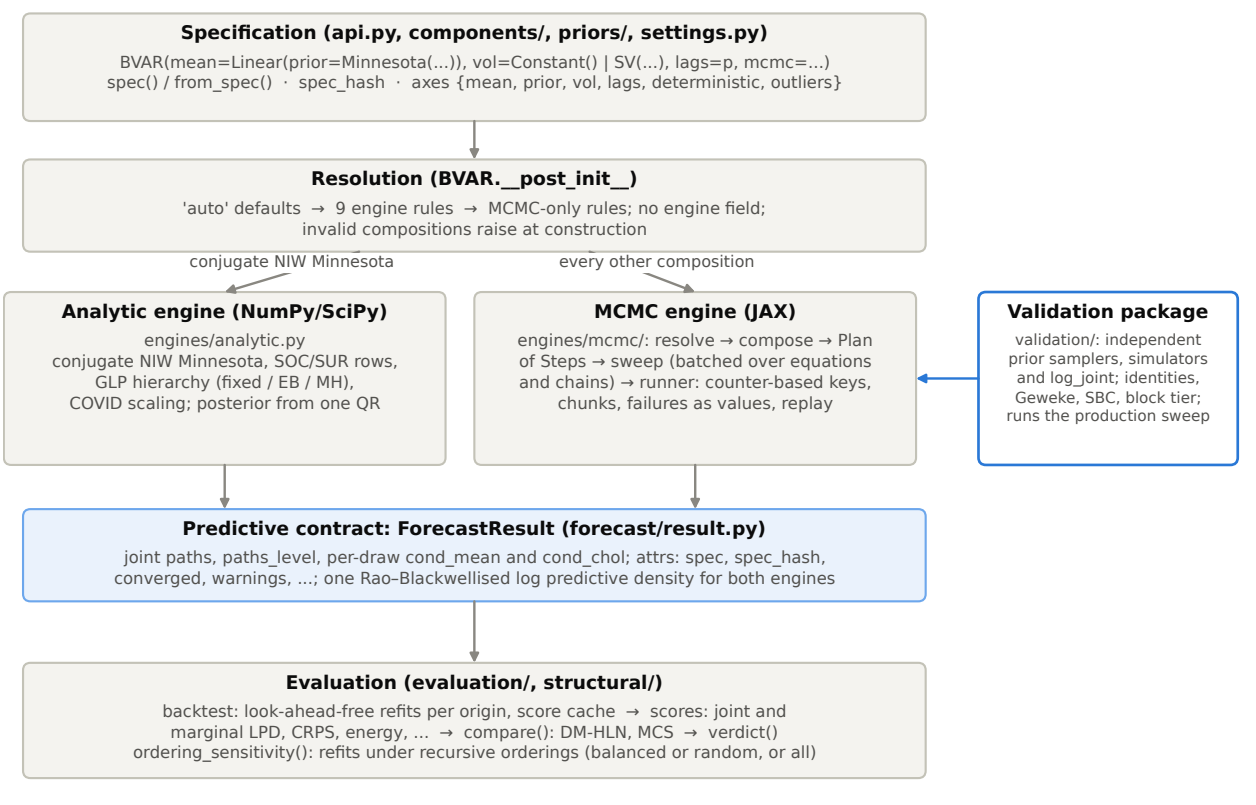
Specification (api.py, components/, priors/, settings.py)
BVAR(mean=Linear(prior=Minnesota(...)), vol=Constant() | SV(...), lags=p, mcmc=...)
spec() / from_spec() · spec_hash · axes {mean, prior, vol, lags, deterministic, outliers}
Resolution (BVAR.__post_init__)
'auto' defaults → 9 engine rules → MCMC-only rules; no engine field;
invalid compositions raise at construction
conjugate NIW Minnesota
every other composition
Analytic engine (NumPy/SciPy)
engines/analytic.py
conjugate NIW Minnesota, SOC/SUR rows,
GLP hierarchy (fixed / EB / MH),
COVID scaling; posterior from one QR
MCMC engine (JAX)
engines/mcmc/: resolve → compose → Plan
of Steps → sweep (batched over equations
and chains) → runner: counter-based keys,
chunks, failures as values, replay
Validation package
validation/: independent
prior samplers, simulators
and log_joint; identities,
Geweke, SBC, block tier;
runs the production sweep
Predictive contract: ForecastResult (forecast/result.py)
joint paths, paths_level, per-draw cond_mean and cond_chol; attrs: spec, spec_hash,
converged, warnings, ...; one Rao–Blackwellised log predictive density for both engines
Evaluation (evaluation/, structural/)
backtest: look-ahead-free refits per origin, score cache → scores: joint and
marginal LPD, CRPS, energy, ... → compare(): DM-HLN, MCS → verdict()
ordering_sensitivity(): refits under recursive orderings (balanced or random, or all)